

CMPE – 211

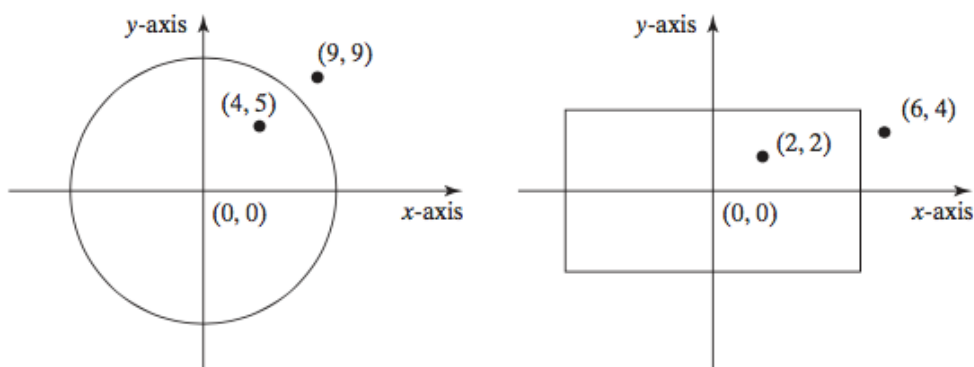
Preliminary Work (Pre-Lab Activity)

Laboratory Experiment #2

Textbook Material: Chapter 3 «Selections»
Chapter 4 «Mathematical Functions, Characters and Strings»
Chapter 5 «Loops»

• • • TASK 1

Write, Compile and Execute a Java program that prompts the user to enter a point (x, y) and checks whether the point is within the circle centered at $(0, 0)$ with radius 10 . For example, $(4, 5)$ is inside the circle and $(9, 9)$ is outside the circle, as shown in Figure below. (Hint: A point is in the circle if its distance to $(0, 0)$ is less than or equal to 10 . The formula for computing the distance is the square root of $((x_2 - x_1)^2 + (y_2 - y_1)^2)$. Test your program to cover all cases.)



Two sample runs are shown below.

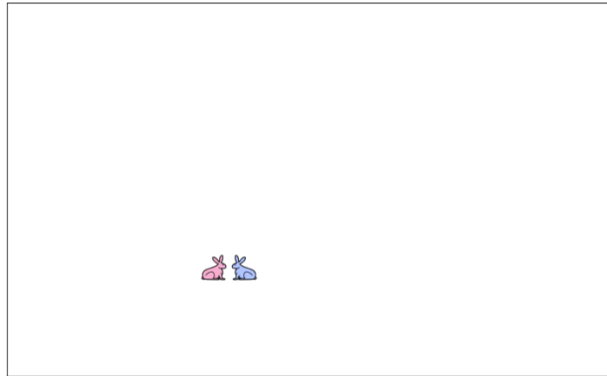
```
Enter a point with two coordinates: 2 2 Enter
Point (2.0, 2.0) is in the rectangle
```

```
Enter a point with two coordinates: 6 4 Enter
Point (6.0, 4.0) is not in the rectangle
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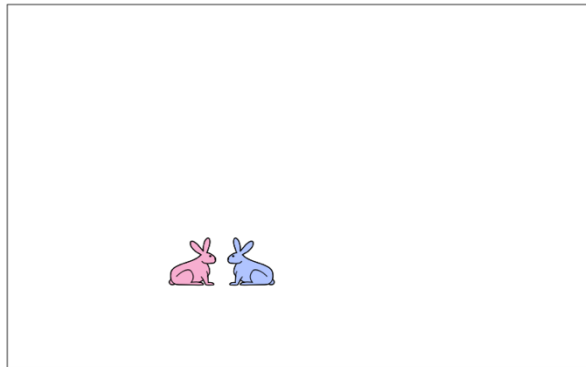
• • • TASK 2

A problem that was defined in the 13th century can be summarized as follows. A newborn pair of rabbits (one male and one female) are put onto an island. Rabbits mate at age of one month and they have a one month gestation period (the time interval each rabbit spends inside the uterus of its mother). We assume that the rabbits do not die and, from second month on, each female produces a one new male and one new female rabbit. The following figure shows the number of rabbits on the island after 4 months.

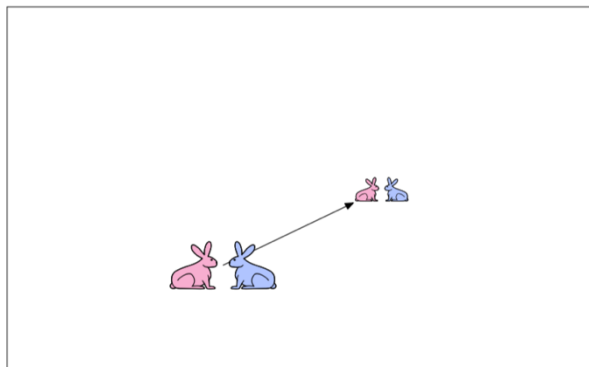
FIRST DAY:



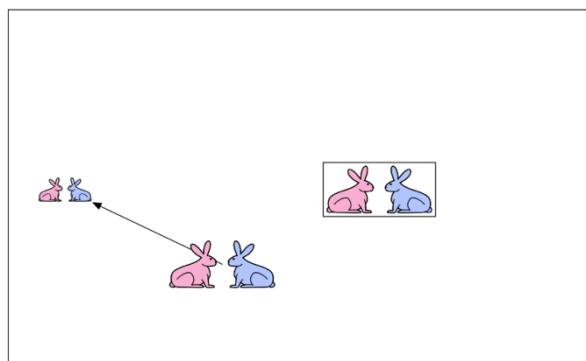
AT THE END OF FIRST MONTH: BECAME MATURE TO MATE:



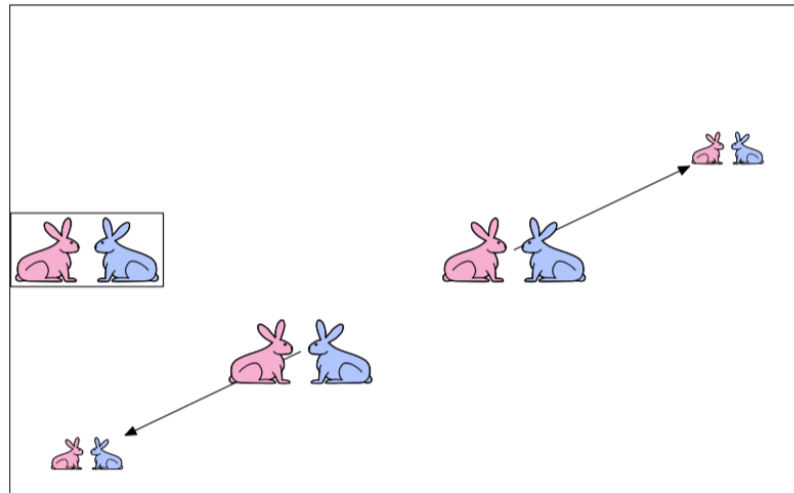
AT THE END OF SECOND MONTH



AT THE END OF THIRD MONTH

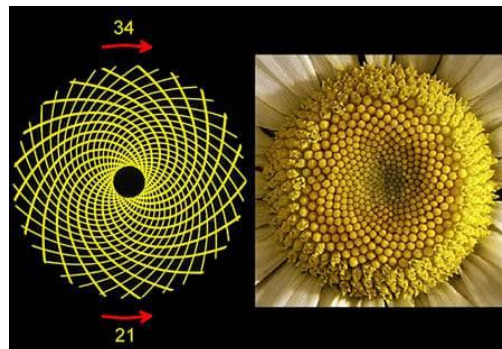


AT THE END OF FOURTH MONTH



Write, Compile and Execute a **Java program** that computes the number of rabbit pairs on the island at the end of 24 months. What is the name of the mathematician who defined the problem?

Notice that the patterns of consecutive numbers obtained are also seen in the nature. For instance, you will find that, after some months, 21 pairs will be available. The next month, there will be 34 rabbit pairs. In fact, these are the numbers of seeds found in a sunflower when counted from left to right and right to left.



• • • TASK 3

The Persian mathematician Al-Karaji (953–1029) computed the first description of Pascal's triangle given below. It was later repeated by the Persian poet-astronomer-mathematician Omar Khayyám (1048–1131); thus the triangle is also referred to as the Khayyam triangle in Iran.

1
1 1
1 2 1
1 3 3 1
1 4 6 4 1
1 5 10 10 5 1
1 6 15 20 15 6 1
1 7 21 35 35 21 7 1

The Pascal's triangle determines the coefficients which arise in binomial expressions. For example, consider the expansion

$$(x + y)^2 = x^2 + 2xy + y^2 = 1x^2y^0 + 2x^1y^1 + 1x^0y^2.$$

The coefficients of the terms are in the third row of the triangle. As a general approach, In general, when $x + y$ is raised to a positive integer power we have:

$$(x + y)^n = a_0x^n + a_1x^{n-1}y + a_2x^{n-2}y^2 + \dots + a_{n-1}xy^{n-1} + a_ny^n,$$

where the coefficients a_i in this expansion are precisely the numbers on row $(n+1)$ of Pascal's triangle. Each row of the triangle can be computed by adding the pairs of numbers in the previous row from left to right. [Write, Compile and Execute](#) a [Java program](#) that prints the Pascal's triangle given above.

• • • TASK 4

[Write, Compile and Execute](#) a [Java program](#) based on a user-defined function that computes the sum of first and last digits of a given number that is composed of two or more digits (e.g. [23](#), [-145](#), [3900](#), etc.).

• • • Sources

- Introduction to Java Programming and Data Structures, Comprehensive Version, 11th Edition, Y. D. Liang, Pearson, 2019.